

Online Examination Report

Date:	22.07.2014
Attempts:	1
Examinee:	Mbuya 2, Denis
Date of birth:	11/10/1986
Subject:	040-Human performance
Result:	72,5 %
Time used:	00:43:36 (HH:MM:SS)

Attachments

1. List of result by question
2. List of result by topic
3. Detailed question result report

Attachment 1: List of result by question

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
1	181	40.	HUMAN PERFORMANCE	0,00	1,00
2	166	40.	HUMAN PERFORMANCE	1,00	1,00
3	175	40.	HUMAN PERFORMANCE	0,00	1,00
4	207	40.	HUMAN PERFORMANCE	1,00	1,00
5	197	40.	HUMAN PERFORMANCE	0,00	1,00
6	162	40.	HUMAN PERFORMANCE	1,00	1,00
7	205	40.	HUMAN PERFORMANCE	1,00	1,00
8	131	40.	HUMAN PERFORMANCE	0,00	1,00
9	20	40.1.2	Accident statistics	0,00	1,00
10	104	40.2	Basic aviation physiology and health maintenance	1,00	1,00
11	106	40.2	Basic aviation physiology and health maintenance	1,00	1,00
12	103	40.2	Basic aviation physiology and health maintenance	1,00	1,00
13	55	40.2.1.1	The atmosphere	1,00	1,00
14	61	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
15	71	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
16	100	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
17	24	40.2.1.2	Respiratory and circulatory systems	0,00	1,00
18	67	40.2.1.2	Respiratory and circulatory systems	1,00	1,00
19	64	40.2.2	Man and Environment: the sensory system	1,00	1,00
20	98	40.2.2.2	Vision	0,00	1,00
21	50	40.2.2.2	Vision	0,00	1,00
22	62	40.2.2.3	Hearing	1,00	1,00
23	22	40.2.2.4	Equilibrium	1,00	1,00
24	8	40.2.2.5	Integration of sensory inputs	1,00	1,00
25	85	40.2.2.5	Integration of sensory inputs	1,00	1,00
26	30	40.2.3	Health and hygiene	0,00	1,00
27	35	40.2.3.2	Body rythm and sleep	1,00	1,00
28	4	40.2.3.3	Problem areas for pilots	1,00	1,00

CQB Number	Internal Number	Syllabus reference	Chapter	Points	Max. Points
29	66	40.2.3.3	Problem areas for pilots	0,00	1,00
30	28	40.2.3.3	Problem areas for pilots	1,00	1,00
31	72	40.2.3.4	Intoxication	1,00	1,00
32	122	40.3	BASIC AVIATION PSYCHOLOGY	1,00	1,00
33	135	40.3.1	Human information processing	0,00	1,00
34	97	40.3.1.1	Attention and vigilance	1,00	1,00
35	80	40.3.1.3	Memory	1,00	1,00
36	40	40.3.2.1	Reliability of human behaviour	1,00	1,00
37	5	40.3.4.3	Co-operation	1,00	1,00
38	114	40.3.5	Human behaviour	1,00	1,00
39	116	40.3.5	Human behaviour	1,00	1,00
40	48	40.3.6.2	Stress	1,00	1,00
Total:72%				29,00	40,00

Attachment 2: List of result by topic

Licence

Date: 22.07.2014

Examinee: Mbuya 2, Denis

Location: Uganda Civil Aviation Authority

Your Result: 29,00 from 40,00 Points (72,50 %)

Topics	Ammount of Questions	Points achieved	Maximum Points	Percent
40. HUMAN PERFORMANCE	40	29,00	40,00	72,50
40.1. Human Factors: basic concepts	1	0,00	1,00	0,00
40.2. Basic aviation physiology and health maintenance	22	17,00	22,00	77,27
40.3. BASIC AVIATION PSYCHOLOGY	9	8,00	9,00	88,89
Total	40	29,00	40,00	72,50

1

In the aeronautical decision making (ADM) process, what is the first step in neutralizing a hazardous attitude?

☒ **Recognizing the invulnerability of the situation.**

☐ **Making a rational judgement.**

☐ **Recognizing hazardous thoughts.**

2

How can smoking affect a pilot?

- ☐ Creates additional carbon dioxide gases in the body which often leads to hyperventilation.
- ☒ Reduces the oxygen-carrying capability of the blood.
- ☐ Can decrease night vision by up to 50 percent.

3

Examples of classic behavioral traps that experienced pilots may fall into are to

- ☐ **complete a flight as planned, please passengers, meet schedules, and 'get the job done.'**
- ☒ **assume additional responsibilities and assert PIC authority.**
- ☐ **promote situational awareness and then necessary changes in behavior.**

4

Which use of cockpit lighting is correct for night flight?

- ☐ Coloration shown on maps is least affected by the use of direct red lighting.
- ☐ The use of regular white light, such as a flashlight, will not impair night adaptation.
- ☒ Reducing the interior lighting intensity to a minimum level.

5

An abrupt change from climb to straight-and-level flight can create the illusion of

☒ **a descent with the wings level.**

☐ **a noseup attitude.**

☐ **tumbling backwards.**

6

What physical change would most likely occur to occupants of an unpressurized aircraft flying above 15,000 feet without supplemental oxygen?

- ☐ Gases trapped in the body contract and prevent nitrogen from escaping the bloodstream.
- ☒ A blue coloration of the lips and fingernails develop along with tunnel vision.
- ☐ The pressure in the middle ear becomes less than the atmospheric pressure in the cabin.

7

Which procedure is recommended to prevent or overcome spatial disorientation?

- ☐ Rely on the kinesthetic sense.
- ☒ Rely on the indications of the flight instruments.
- ☐ Reduce head and eye movements to the extent possible.

8

The Decide Model is comprised of a 6-step process to provide a pilot a logical way of approaching Aeronautical Decision Making. These steps are:

- ☒ Detect, estimate, choose, identify, do, and evaluate.
- ☐ Determine, eliminate, choose, identify, detect, and evaluate.
- ☒ Determine, evaluate, choose, identify, do, and eliminate.

9

The distribution of primary causes of accidents in the worldwide jet aircraft commercial fleet shows that human error is involved in:

- ☐ about 90% of cases
- ☐ all cases, one way or another
- ☒ 70% of cases
- ☐ about 70% of cases

10

Which of these would most likely result in hyperventilation?

☐ Excessive carbon monoxide.

☒ Insufficient carbon dioxide.

☐ Insufficient oxygen.

11

To overcome the symptoms of hyperventilation, a pilot should

☐ increase the breathing rate.

☒ slow the breathing rate.

☐ swallow or yawn.

12

Which of these is a common symptom of hyperventilation?

☐ Decreased breathing rate.

☒ Drowsiness.

☐ A sense of well-being.

13

Air at an altitude of 18.000 feet contains, approximately:

☒ **21% oxygen**

☐ **15% oxygen**

☐ **10% oxygen**

☐ **5% oxygen**

14

Flying immediately after SCUBA diving involves the risk of getting:

- ☐ hypoxia
- ☐ hyperventilation
- ☒ decompression sickness
- ☐ stress

15

When exhaling, the expired air contains:

- ☐ more nitrogen than in the inhaled air
- ☐ more oxygen than in the inhaled air
- ☐ less water vapour than in the inhaled air
- ☒ more carbon dioxide than in the inhaled air

16

Positive G will cause the blood flow to the brain to:

☐ first increase, then decrease

☒ decrease

☐ increase

☐ remain constant

17

When faced with sustained cold temperature, how does the body resist this physical stress?

- ☒ **By vasodilatation which permits a greater flow of blood to the periphery.**
- ☐ **By increasing cardiac frequency.**
- ☐ **By speeding up the metabolic rate in the Autonomic Nervous System.**
- ☐ **By intense vasoconstriction.**

18

Flying immediately following a dive with SCUBA diving equipment to a depth greater than 10 metres

- ☐ will always lead to hypoxia
- ☒ can cause decompression sickness even when flying at pressure altitudes below 18 000 FT
- ☐ has no influence on altitude flights
- ☐ presents any dangers caused by DCS (decompression sickness) when climbing to altitudes not exceeding 30 000 FT

19

Perceptual illusions are:

- ☐ usually not serious except in bad weather.
- ☐ never caused by the brain's ability to distort normal sense inputs.
- ☐ abnormal and indicate weakness in one or more of the senses.
- ☒ normal and can be prevented by trusting instrument read-out.

20

The retina allows for colour perception as a result of the:

- ☐ rods located in its peripheral zone
- ☒ cones located in its central part
- ☐ crystalline lens
- ☒ rods located in its central part

21

Cataract is caused by:

- ☐ Lack of mobility of the cornea
- ☐ A mis-shapened cornea
- ☐ A clouding of the lens
- ☒ A lack of accommodation at the cornea

22

Which part of the inner ear is responsible for the perception of sound?

- ☐ The semicircular canals
- ☐ The Eustachian tube
- ☒ The cochlea
- ☐ The sacculus and utriculus

23

Through which part of the ear does the equalization of pressure take place, when altitude is changed?

- ☐ Cochlea
- ☐ External auditory canal
- ☒ Eustachian tube
- ☐ Tympanic membrane

24

Which sensations does a pilot get, when he is rolling out of a prolonged level turn?

- ☐ Flying straight and level
- ☐ Turning into the original direction
- ☒ Turning in the opposite direction
- ☐ Climbing

25

"Pilot's vertigo"

- ☐ announces the beginning of airsickness
- ☒ is the condition of dizziness and/or tumbling sensation caused by contradictory impulses to the central nervous system (CNS)
- ☐ is the sensation of climbing caused by a strong linear acceleration
- ☐ is the sensation to keep a rotation after completing a turn

26

The following can be observed when the internal body temperature falls below 35°C:

- ☒ the appearance of intense shivering
- ☐ profuse sweating
- ☐ mental disorders, and even coma
- ☐ shivering, will tend to cease, and be followed by the onset of apathy

27

The physiological rhythms of a pilot in a new time zone will resynchronise to this new time zone at a rate of about

- ☒ 1 - 1.5 hours a day
- ☐ 3 - 3.5 hours a day
- ☐ 4 - 4.5 hours a day
- ☐ 2 - 2.5 hours a day

28

How is Yellow fever contracted?

- ☐ By contact with the saliva of infected animals.
- ☒ A virus transmitted by an infected mosquito.
- ☐ Contaminated food or water.
- ☐ Excessive use of alcohol.

29

Loss of elasticity of the crystalline lens due to ageing occurs in:

- ☒ presbyopia
- ☐ hypermetropia
- ☐ astigmatism
- ☒ myopia

30

Which of the following statements about hyperthermia is correct ?

- ☐ Vasodilatation is the only mechanism which is capable of reducing body temperature.
- ☐ Performance is not impaired by an increase in body temperature to 40°C or more.
- ☒ Complete adaptation to the heat in a hot country takes about a fortnight.
- ☐ Evaporation is more effective when ambient humidity is high.

31

Alcohol degrades:

- ☐ Slow wave sleep
- ☐ Stage 3 and 4 sleep
- ☒ Paradoxical sleep
- ☐ Stage 2 sleep

32

What does good cockpit stress management begin with?

- ☐ Eliminating life and cockpit stress issues.
- ☒ Good life stress management.
- ☐ Knowing what causes stress.

33

While material is being taught, students may be learning other things as well. What is the additional learning called?

- ☐ Residual learning.
- ☐ Incidental learning.
- ☒ Conceptual learning.

34

Which of the following statements are correct?

1. The first information received determines how subsequent information will be evaluated.
2. If one has made up one's mind, contradictory information may not get the attention it really needs.
3. With increasing stress, attention is limited thereby reducing the flow of information to the central decision maker.

- ☐ 1 and 3 are correct
- ☐ 2 and 3 are correct
- ☐ 1 and 2 are correct
- ☒ 1, 2 and 3 are correct

35

Working memory enables us, for example,

- ☒ to remember a clearance long enough to write it down
- ☐ to ignore messages for other aircraft
- ☐ to remember our own name
- ☐ to store a large amount of visual information for about 0.5 seconds

36

Situations particularly vulnerable to "reversion to an earlier behaviour pattern" are:

1. when concentration on a particular task is relaxed
2. when situations are characterised by medium workload
3. when situations are characterised by stress

☐ 1. and 2.

☐ 2. and 3.

☐ 3 only.

☒ 1. and 3.

37

Which of the following statements best characterises a self-centered cockpit ?

- ☐ The egoistic and self-centered personality of the captain often leads to a synergetic cockpit
- ☐ While decreasing communication, the independence of each member bolsters the crew's synergy
- ☒ Each one does his own thing while at the same time assuming that everyone is aware of what is being done or what is going on
- ☐ The communication between crew members always increases when the captain takes charge of a situation

38

Some of these dangerous tendencies or behavior patterns which must be identified and eliminated include:

- ☐ Performance deficiencies from human factors such as, fatigue, illness or emotional problems.
- ☒ Peer pressure, get-there-itis, loss of positional or situation awareness, and operating without adequate fuel reserves.
- ☐ Deficiencies in instrument skills and knowledge of aircraft systems or limitations.

39

Hazardous attitudes which contribute to poor pilot judgment can be effectively counteracted by

- ☐ taking meaningful steps to be more assertive with attitudes.
- ☐ early recognition of hazardous thoughts.
- ☒ redirecting that hazardous attitude so that appropriate action can be taken.

40

The behavioural effects of stress may include:

- 1: manifestation of aggressiveness.**
- 2: a willingness to improve communication.**
- 3: a willingness for group cohesion.**
- 4: a tendency to withdrawal.**

The combination of correct statements is:

- ☐ **3 and 4 are correct**
- ☐ **2,3 and 4 are correct**
- ☒ **1 and 4 are correct**
- ☐ **1,2 and 3 are correct**